Xingyu Chen

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Education

2023 – present **Shanghai Jiao Tong University**, Minhang, Shanghai 200240, China
Ph.D. Candidate in Statistics
Advisor: **Lin Liu**

2019 – 2023 **Huazhong University of Science and Technology**, Hongshan, Wuhan 430074, China
B.S. in Mathematics and Applied Mathematics

Research Interests

High-dimensional statistics; Causal inference; Semiparametric theory; Nonparametric theory.

Publications

[1] **Xingyu Chen**, Lin Liu, and Ruiqi Zhang. On computing and the complexity of computing higher-order U-statistics, exactly. *Statistics and Computing*, 36(5), 2026. Article 223. doi:10.1007/s11222-026-10974-x.

Preprints

[1] **Xingyu Chen**, Lin Liu, and Rajarshi Mukherjee. Method-of-moments inference for GLMs and doubly robust functionals under proportional asymptotics. *arXiv preprint arXiv:2408.06103, 2024*.

Software

2025, 2026 `u-stats` (Python package) and `ustats` (R package)– Computing Higher-order U-statistics. Available on PyPI / CRAN.

2026 `HOIF` (R package) –Computing Higher-Order Influence Function (HOIF) estimators for the average treatment effect (ATE). Available on CRAN.

Awards

Honors and Awards during Ph.D.

2026 Mitacs Globalink Research Award (GRA), Mitacs, Canada
Visiting research internship, University of Toronto, 24 weeks in 2027
Host supervisor: Prof. Linbo Wang; home supervisor: Prof. Lin Liu

May 2025 Best Paper Award
Forum on AI for Mathematics, Shanghai Jiao Tong University, Shanghai, China

Apr. 2025 Travel Award
3rd Annual Meeting of the Chinese Statistical Association of Young Scholars, Jinan, China

Honors and Awards during Undergraduate

Jun. 2023 Outstanding Graduate
Huazhong University of Science and Technology

Dec. 2021 First Prize
13th National Undergraduate Mathematics Competition (Hubei Division)

Dec. 2020 National Scholarship for Undergraduate Students
Ministry of Education of China

Presentations

May 2026 Oral, The 2nd Science and Technology Academic Month Doctoral Forum, School of Mathematics and Statistics, Huazhong University of Science and Technology, Wuhan, China

Jan. 2026 Poster, 2026 Joint Workshop of RMTA and SNAB[1], Nanjing, China

Nov. 2025 Oral, The 1st National High-level Doctoral Student Academic Forum, Shanghai Jiao Tong University, Shanghai, China

Nov. 2025 Oral, The 6th Beijing-Tianjin-Hebei Workshop on Mathematics and Statistics & Graduate Student Forum, Baoding, China

Nov. 2025 Oral, The 3rd National Frontier Mathematics Doctoral Student Academic Exchange Forum, Peking University, Beijing, China

Sep. 2025 Oral, 2025 Joint Annual Conference of CAAS-SIRB and CAISER-ESDSB[2], Shanghai, China

May 2025 Oral, Forum on AI for Mathematics, Shanghai Jiao Tong University, Shanghai, China

Apr. 2025 Oral, 3rd Annual Meeting of the Chinese Statistical Association of Young Scholars, Jinan, China

Apr. 2025 Poster, European Causal Inference Meeting 2025, Ghent, Belgium

[1] RMTA: Random Matrix Theory and Applications; SNAB: Statistical Network Analysis and Beyond.
[2] CAAS-SIRB: Statistical Interdisciplinary Research Branch of Chinese Association for Applied Statistics; CAISER-ESDSB: Ethnic Statistics and Data Science Branch of Chinese Association of Industrial Statistical Education and Research.

Service

Dec. 2022 Main contributor of the Chinese translation of "Fernholz, L. T., and S. Morgenthaler. A conversation with John W. Tukey and Elizabeth Tukey. Statistical Science, 15(1): 79–94, 2000." for Capital of Statistics (COS), a leading online platform for the Chinese statistical community, established in 2006.

Teaching Experience

Fall 2025 Teaching Assistant
Advanced Mathematical Statistics (Ph.D. Course)
School of Mathematical Sciences, Shanghai Jiao Tong University
Instructor: Lin Liu

Spring 2025 Teaching Assistant
Probability and Statistics (Undergraduate Course)
School of Mathematical Sciences, Shanghai Jiao Tong University
Instructor: Peng Luo

Fall 2024 Teaching Assistant
Linear Algebra (Undergraduate Course)
School of Mathematical Sciences, Shanghai Jiao Tong University
Instructor: Zhen Cui

Computation Skills

R, Python, Mathematica, C++, Matlab